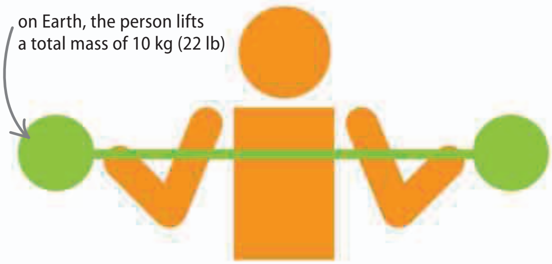
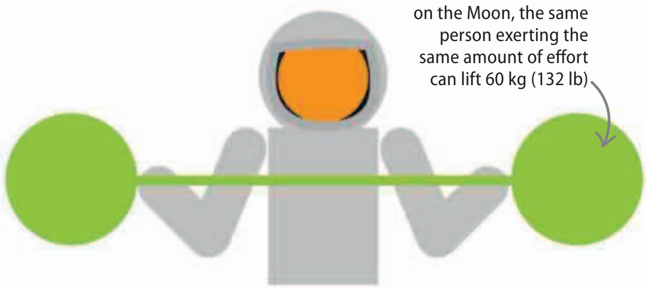
Weight and mass

Weight is not the same thing as mass. Mass is the amount of matter an object contains, while weight is the force with which the Earth or another body pulls on the object. An object can have the same mass but weigh differently, depending on the gravitational force acting on the object.

Someone who weighs **40 kg (88 lb) on Earth** would weigh **more than one tonne on the Sun**—if it were possible to stand on its surface.



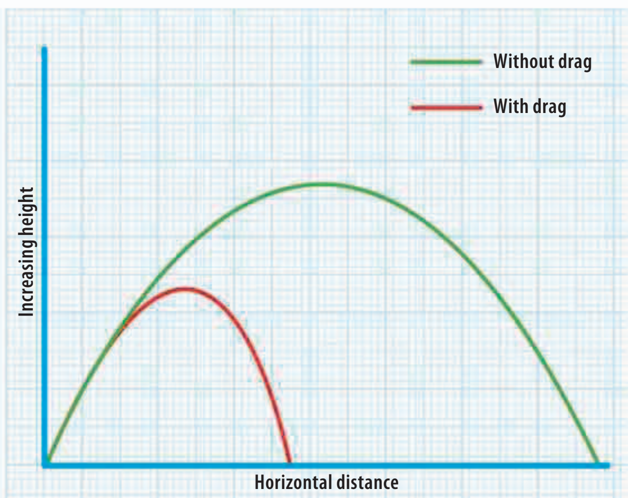
△ **On Earth**
Lifting a barbell requires this person to exert a greater force than the barbell's weight. Here, a weightlifter is lifting a barbell with weights whose total mass is 10 kg (22 lb).



△ **On the Moon**
On the Moon, the force of gravity is about one sixth that on Earth. So the same effort is required to lift 60 kg (132 lb) on the Moon as it is to lift 10 kg (22 lb) on Earth.

Ballistics

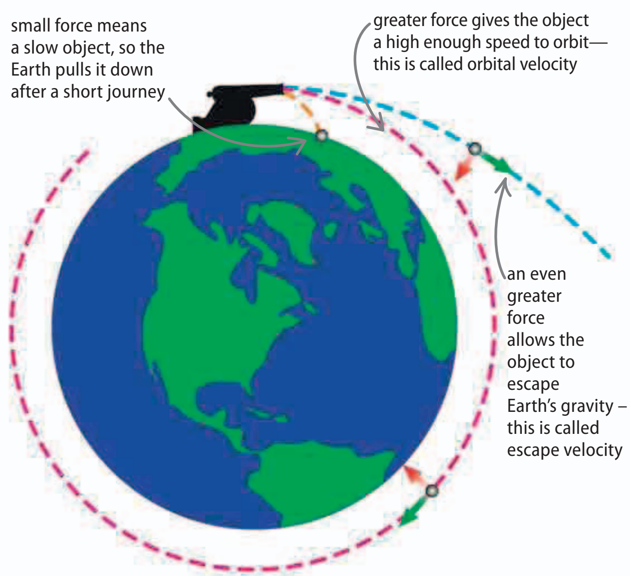
A thrown object is pulled back down to Earth by the force of gravity. At the same time, its sideways motion is decelerated by the drag force applied by the air. If there were no atmosphere, the object would travel a much greater distance.



△ **Air resistance and motion**
On Earth, air resistance applies a drag force, slowing the object's motion (red line). Where there is no air resistance, such as on the Moon, a thrown object moves in a parabola (green line)—a steady speed in a horizontal direction, while moving up and then down.

Orbit

The harder an object is thrown, the faster and farther it travels before its path takes it back to the Earth's surface. If the object is propelled hard enough, it will gain enough speed to counteract the pull of gravity so it will never land and will orbit the Earth.



△ **Newton's satellite**
This diagram is based on an illustration by Isaac Newton. He showed that if a cannonball is fired with enough force, its speed will allow it to orbit or escape Earth completely.